

HAMMER PROCESS ROUTER

This hammer exercise is designed to provide students with practice on the various machines in the machine shop. Start work early and take pride in your work. Always stay safe, ask questions, and have fun!

The hammer cutting bit boxes contain: 90° countersink tool, 13/32" endmill, 7/16" endmill, 13/32" drill bit, 15/32" drill bit, and a #7 (0.201") drill bit.

Head

(Dwg No. 1/5, Rev 01)

Process	Machine/Tool	Comments
Face one end of the stock	Lathe, right hand HSS cutting tool	Once the stock is set up in the lathe, complete all the features on that side before taking it out of the lathe. Leave only about 0.5"-0.75" sticking out of the collet. Your facing pass should be shallow.
Reduce end diameter	Lathe, right hand HSS cutting tool	To reduce the shoulder diameter, take a measurement cut, at least three roughing cuts, then a finishing cut to spec. Keep checking the diameter with your calipers!
Drill hole for soft face	Lathe, center drill, drill bit	Use a center drill to start the hole, then drill to 1.200". Use cutting fluid for all operations. Practice peck drilling—back out often to break up chips.
Produce countersink	Lathe, right hand carbide cutting tool, lathe compound	Angle the lathe compound and toolpost to approximately the 75° compound marking. Move the cutting tool close to the part in z-direction and leave it there. Using the compound wheel, take small cuts to produce the countersink, moving in the x-direction to achieve the desired radius. Remember to run the lathe at 1200 RPM. <i>Talk to a proctor before starting.</i>
Tap hole for soft face	Lathe, tap (w/handle and guide)	Use cutting fluid. Turn the tap back frequently to break up chips and cut clean threads.
Face other end to length	Lathe, right hand HSS cutting tool	Make a shallow facing pass on the other end of the stock. Take the piece out and measure the length. Touch off to zero the z-axis, then face to length.
Reduce end diameter	Lathe, right hand HSS cutting tool	Again, make a measurement cut, three roughing cuts, and a finishing cut to spec.
Drill hole for hard face	Lathe, center drill, drill bit	Start a pilot hole with the center drill. Drill using the machine length stub drill that is 1/32" undersize. Use cutting fluid and practice peck drilling. Check your speeds!
Countersink hole for hard face	Lathe, 90° countersink tool	This step will help you fit your hard face into the hammer head. Use 45-45-90 triangles to figure out how far to feed in the z-direction. Use cutting fluid and check your speeds for this operation.

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Ream hole to size	Lathe, reamer	Be careful not to ream to the bottom of the hole. If the reamer hits the bottom of the hole, it can damage the reamer and your part. Use cutting fluid for this operation. If the reamer chatters significantly, decrease the RPM.
Cut flats	Mill, end mill, fly cutter	Fixturing: Place your hammer head on a double thick parallel with the flat ends against the jaws of the vise. <i>Note: It is recommended to tighten the vise extra tightly with the mallet because it is common for the head to come loose during this step.</i> Roughing cut: Using the 7/16" end mill at a speed of 400 RPM and depth of cut of 0.06", power feed using a feed rate of 6 in/min. Remove material in two parallel passes such that the stepover is about half the diameter of the end mill. Finishing cut: Using the fly cutter with a speed of 800 RPM and power feed rate of 3 in/min, make a finishing cut of depth 0.01". Take your hammer head out and measure how much you've milled off. Then, flip the hammer head and repeat this step to mill the other flat so that the distance between the flats is 0.860". Adjust the depth of the roughing cut as necessary, but leave the finishing cut at 0.01".
Drill hole for slot	Mill, drill chuck, center drill, drill bit	Determine desired x-coordinate from the drawing (hint: use the radius of the slot). Drill a single hole at one end of the slot, using a center drill to create a pilot hole. Use a 1/32" undersized drill bit and cutting fluid for this operation. Practice peck drilling, and don't use the thick parallel.
Mill slot to spec	Mill, undersize end mill, full-size end mill	Plunge milling: Use an undersize end mill to plunge overlapping thru-holes from one slot centerpoint to the other, starting at the drilled hole. Each plunge pass should be shifted over by 0.075" to 0.1". Roughing slot: Make transverse cuts at increasing z depths using the undersized end mill until it is through the part. Finishing slot: Repeat the same series of transverse cuts, this time with the full-size end mill.
Deburr	File	Deburr the edges of the milled slot using a rat-tail file.
Stamp E4 number	Number stamps, arbor press	Try on a piece of scrap first! Arbor press: Make sure the number stamp is as vertical as possible. You do not need to use significant force.

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Hard Face

(Dwg No. 3/5, Rev 01)

Process	Machine/Tool	Comments
Face end	Lathe, right hand carbide cutting tool	Leave time to finish all lathe operations on a single side before removing the part from the collet. Remember to check your speeds for all operations on the hard face.
Reduce large diameter	Lathe, right hand carbide cutting tool	Reduce the diameter for a length of 1" along the z-axis. Roughing cuts: Each pass should not exceed 0.030" deep. Finishing and measurement cuts: Make a final pass with a slow feed rate to produce a smooth and accurate finish. These passes should be around 0.010" deep. Use autofeed and keep checking the diameter! <i>Talk to a proctor about autofeed.</i>
Measure hole for hard face	Hammer head, pin gauges	To make sure your hard face press fits into your hammer head, use the pin gauges to measure the hammer head hole for the hard face. Adjust the small diameter of your hard face accordingly. <i>Talk to a proctor about using the pin gauges.</i>
Reduce small diameter	Lathe, right hand carbide cutting tool	Roughing cuts: Each pass should not exceed 0.030" deep. Finishing cut: Make a final pass with a slow feed rate to produce a smooth and accurate finish. This pass should be around 0.010" deep. This diameter is extremely important. Remember, this needs to press fit into your hammer head. Use autofeed and keep checking the diameter!
Chamfer	Lathe, 45° HSS cutting tool	Use the 45° cutting tool and feed in either the x- or z-direction. Again, use 45-45-90 triangle properties to calculate how much to remove.
Cut part from stock	Horizontal bandsaw	Cut your part off from the bar stock using the horizontal bandsaw. Make sure you cut off at least 1/8" more than the length spec!
Optional: Stamp E4 number on stem	Number stamps, arbor press	To facilitate identification of your hard face in later steps such as heat treating. Try on a piece of scrap first! Arbor press: Make sure the number stamp is as vertical as possible. You do not need to use significant force.

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Fixture part on lathe by holding the small diameter in collet	Lathe	Make sure the collet is in contact with the cylindrical small diameter, not the small radius where the small and large diameters meet.
Face to length	Lathe, right hand carbide cutting tool	First, make a measurement cut just deep enough to remove the saw marks. Remove the part and measure its length to see how much material to face off. Leave 0.005"-0.01" extra for the upcoming rounding operation.
Create rounded end	Lathe, pneumatic grinder with 80-grit disk, file	Remove the cutting tool and anything sharp in the tailstock. Use the pneumatic grinder to grind the radius to spec. Use the aluminum radius gauge to check that you have achieved the specified curvature. Run the lathe at 800 RPM.
Polish	Lathe, sandpaper or pneumatic grinder with polishing disk	Get sandpaper or a polishing disk from a proctor. Sandpaper: Let the sandpaper do the work. DO NOT wrap the sandpaper around the part.
Heat-treat hard face	Heat-treating ovens	Locate the gloves, fire extinguisher, quenching oil, and heat-treating instructions before turning the oven on. <i>Follow the heat-treating instructions for this step.</i>
Sandblast	Sandblasting cabinet	This will remove the carbon soot without changing the dimensions of your part. Sandblast until your hard face has a uniform matte gray-silver surface finish.
Press-fit into hammer head	Arbor press	Ensure the hard face is vertical. If it is misaligned, it will seize up and ruin the main body.

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Handle

(Dwg No. 4/5, Rev 01)

Process	Machine/Tool	Comments
Engrave wood blank	Laser cutter	The engraving will show you where to cut out the rough shape of the hammer handle. <i>Refer to the Hammer Handle Laser Etching Instructions on the website>E4>Hammer page.</i>
Cut the first contour	Vertical bandsaw	Cut around the laser-cut contour, leaving a little extra for sanding. Make relief cuts so that the bandsaw blade doesn't have to make sharp turns.
Sand the first perimeter	Spindle sander, belt sander	Be careful not to sand too much! You can always sand more later, but you can't un-sand!
Make the tape template	Laser cutter	Put a piece of blue masking tape on a sacrificial wood piece and cut out the tape template file.
Cut the second contour and slot for the wedge	Vertical bandsaw	The slot for the wedge should be just about one bandsaw blade thick. You can widen it later on if your wedge doesn't fit.
Sand the second perimeter	Spindle sander, belt sander	Make sure you don't sand too much; you can always sand more, but you can't un-sand!
Create 3/8" radii	Router table	Use the modified parallel wood clamp for this operation. Since 3/8" radii are the largest radii on the handle, you can radius the whole handle without worrying about cutting too far.
Blend the radii together	Pneumatic grinder	Using the pneumatic grinder, smooth out the handle shape. Practice your technique on scrap wood first and use the radius gauges to check your work.
Hand sand the whole handle	Sandpaper	Start with 80 or 100 grit sandpaper to remove all machining marks, blend in the different radii, and round the 1/8" radius at the end of the handle. Finish sanding with 180 grit sandpaper. Throw your sandpaper away or give it to someone else when you are done, do not leave it sitting around.
Stamp E4 number	Number stamps, hammer	Try on a piece of scrap first! Hammer: Stamp each number with a single strike, otherwise you will get shadows.

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Soft Face

(Dwg No. 2/5, Rev 01)

Process	Machine/Tool	Comments
Face the end of the stock	Lathe, right hand HSS cutting tool	Extend the material about 1.125" beyond the face of the collet so that there is enough extra material for the upcoming operations.
Reduce diameter	Lathe, right hand HSS cutting tool	Reduce the diameter for a length of 0.8-0.85" along the z-axis.
Drill and tap hole	Lathe, drill bit, tap (w/ handle and guide)	Do not tap too far. Back out frequently while drilling and tapping to prevent chip buildup. Use cutting fluid for tapping.
Part off	Lathe, HSS parting tool	Part off enough material to face to length. Remember to run the lathe at 370 RPM.
Face parted end to length	Lathe, right hand HSS cutting tool	Before putting the part in the lathe, measure the length using calipers. Touch off and zero in the z-direction, then face to length.
Chamfer	Lathe, 45° HSS cutting tool	Use the 45° cutting bit and feed in either the x- or z- axis to cut the desired chamfer size. Use 45-45-90 triangle properties to determine how far to feed in the x- or z-direction.